Nishant Balepur

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Research Summary

I am a computer science Ph.D. candidate at the University of Maryland, advised by Professor Jordan Boyd-Graber and Rachel Rudinger. I study Natural Language Processing, but use methods from Machine Learning and Human-Computer Interaction. I develop and deploy collaborative AI systems that measurably help users achieve their goals, with a goal of improving user feedback, personalization, reasoning, factuality, and evaluations beyond just correctness and preferences.

Education

University of Maryland, College Park (UMD) — College Park, MD
Ph.D. Computer Science; GPA: 4.00/4.00 — Aug 2023 - Present
Advisors: Jordan Boyd-Graber, Rachel Rudinger
Proposed Thesis: Teaching AI to Answer Questions with Reasoning that Actually Helps You [PDF]
Committee: Jordan Boyd-Graber, Rachel Rudinger, Shi Feng, Fumeng Yang, David Weintrop

New York University — New York, NY
Visiting Student — Aug 2025 - May 2026
Host: Eunsol Choi

University of Illinois at Urbana-Champaign (UIUC) — Urbana, IL
B.S. Computer Science; B.S. Statistics (Dual Degree); GPA: 4.00/4.00 — Aug 2019 - May 2023

Industry Research Experience

Allen Institute for AI (Ai2) — Seattle, WA
Research Scientist Intern: Asta (Self-Evolving, Overnight, Collaborative Agents) — Sept 2026 - Present

Microsoft Research (MSR) — New York, NY
Research Intern: AI Interaction and Learning (Coding Agent Evaluation) — May 2026 - Aug 2026
Outcome: One paper under review at ICLR'27

Allen Institute for AI (Ai2) — Seattle, WA
Research Scientist Intern: Semantic Scholar (Personalized Deep Research Agents) — May 2025 - May 2026
Outcome: One ACL'26 paper, One paper under review, co-authored ICLR'26 oral, Demo

Adobe — San Jose, CA
Research Scientist Intern: Document Intelligence (Multi-Agent Question Answering) — May 2024 - Aug 2024
Outcome: Paper at NAACL 2025

Meta — Menlo Park, CA
Software Engineering Intern: Facebook Creators Well-being (Comment Safety) — May 2022 - Aug 2022
Outcome: Intervention with 15% drop in negative interactions across Facebook's 1 billion users

Fellowships, Grants, and Scholarships

NSF Graduate Research Fellowship Program (GRFP) — UMD
Funded proposal on NLP for information accessibility, awarding $159,000 over three years — April 2023 - April 2026

Cohere for AI Research Grant (2x Awardee) — UMD
Funded proposals on LLM for Education and AI Safety, receiving full access to Cohere models — April 2024

Dean's Fellowship — UMD
Awarded the Dean's Fellowship for outstanding academic achievement — April 2023 - April 2025

Illinois Engineering Achievement Scholarship (2x Awardee) — UIUC
Merit-based scholarship for undergraduate students — June 2021, 2022

Awards

Certificate of Outstanding Achievement (2x) — UMD
Outstanding performance and exceptional contributions to the CS Graduate Program — April 2025, 2026

MASC-SLL 2025 Best Paper Award — UMD
Best paper award for our MCQA Position Paper by Penn State University — April 2025

MASC-SLL 2024 Best Paper Award — UMD
Best paper award for "Artifacts or Abduction" by Johns Hopkins University — May 2024

Outstanding Reviewer Award (2x Awardee) — UMD
Selected as one of 51 outstanding reviewers for the EMNLP conference out of 10,000+ reviewers — Nov 2024, 2025

Bronze Tablet — UIUC
Name inscribed in the university library for graduating in the top 3% of students in the class of 2023 — May 2023

Computer Science Graduation with Highest Honors — UIUC
Recommended by the UIUC computer science department to graduate with highest honors — May 2023

C.W. Gear Outstanding Undergraduate Student — UIUC
Awarded to two undergraduate CS seniors (fourth year) for excellence in research and service — May 2022

James N. Snyder Memorial Award — UIUC
Awarded to three undergraduate CS juniors (third year) based on academic merit — May 2021

UIUC Datathon Champion — UIUC
Developed the best machine learning model for analyzing consumer behavior out of 150 UIUC students — April 2021

State Farm Programming Competition Finalist — UIUC
Top four finalist out of 50 national teams for solving data analysis questions — Nov 2020

Conference Publications

* denotes equal contribution, † denotes mentored student

19. Check The Scoreboard: An Analysis of Scoring Schemes on Multiple-Choice Evaluation [PDF]
EMNLP 2026
Nishant Balepur, Paiheng Xu, Wei Ai, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber

18. Language Models Don't Know What You Want: Personalization in Deep Research Needs Real Users [PDF]
ACL 2026
Nishant Balepur, Sergey Feldman, ..., Joseph Chee Chang, Eunsol Choi, Jordan Boyd-Graber, Aakanksha Naik
Selected: We deploy a personalized Deep Research system and collect user feedback to uncover novel failures

17. Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers [PDF]
ACL 2026
Nishant Balepur, Atrey Desai†, Rachel Rudinger

16. BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks [PDF]
ACL 2026
Nishant Balepur, Bhavya Rajasekaran†, Jane Oh†, Michael Xie†, Atrey Desai†, ..., Jordan Boyd-Graber

15. Measuring Users' Mental Models of Speech Translation in Human-MT Collaboration [PDF]
ACL 2026, IWSLT 2026 (Oral)
Hyojung Han, **Nishant Balepur**, Jordan Boyd-Graber, Marine Carpuat

14. AstaBench: Rigorous Benchmarking of AI Agents with a Holistic Scientific Research Suite [PDF]
ICLR 2026 (Oral)
Jonathan Bragg*, Mike D'Arcy*, **Nishant Balepur**, ..., Yoav Goldberg, Ashish Sabharwal, Daniel S. Weld

13. A Good Plan is Hard to Find: Aligning Models with Preferences is Misaligned with What Helps Users [PDF]
EMNLP 2025
Nishant Balepur, Matthew Shu†, ..., Shi Feng, Fumeng Yang, Rachel Rudinger, Jordan Boyd-Graber
Selected: We show optimizing on user/LLM judgments and agents simulations cannot capture what helps users

12. Which of These Describes Multiple Choice Evaluations? A) Forced B) Flawed C) Fixable D) All of the Above [PDF]
ACL 2025 (Oral, ACL Best Paper Nomination, Best Paper Award at MASC-SLL 2025)
Nishant Balepur, Rachel Rudinger, Jordan Boyd-Graber
Selected: We expose flaws multiple-choice evaluations and design solutions from education to remedy them

11. Whose Boat Does it Float? Improving Personalization in Preference Tuning via Inferred User Personas [PDF]
ACL 2025
Nishant Balepur, Vishakh Padmakumar, Fumeng Yang, Shi Feng, Rachel Rudinger, Jordan Boyd-Graber

10. Reverse Question Answering: Can an LLM Write a Question so Hard (or Bad) that it Can't Answer? [PDF]
NAACL 2025, Oral
Nishant Balepur, Feng Gu, Abhilasha Ravichander, Shi Feng, Jordan Boyd-Graber, Rachel Rudinger

9. Moderating a Mixture of Document Speakers to Summarize Debatable Queries in Document Collections [PDF]
NAACL 2025, Oral
Nishant Balepur, Alexa Siu, Nedim Lipka, Franck Dernoncourt, Tong Sun, Jordan Boyd-Graber, Puneet Mathur

8. A Smart Mnemonic Sounds like Glue Tonic: Mixing LLMs with Student Feedback to Make Learning Stick [PDF]
EMNLP 2024
Nishant Balepur, Matthew Shu†, Alexander Hoyle, ..., Shi Feng, Seraphina Goldfarb-Tarrant, Jordan Boyd Graber
Selected: We optimize language models directly on student learning outcomes to improve mnemonic generation

7. KARL: Knowledge-Aware Retrieval and Representations aid Retention and Learning in Students [PDF]
EMNLP 2024
Matthew Shu*†, **Nishant Balepur***, Shi Feng*, Jordan Boyd-Graber

6. Plausibly Problematic Questions in Multiple-Choice Benchmarks for Commonsense Reasoning [PDF]
EMNLP 2024 (Findings)
Shramay Palta, **Nishant Balepur**, Peter Rankel, Sarah Wiegreffe, Marine Carpuat, Rachel Rudinger

5. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question? [PDF]
ACL 2024, Best Paper Award at MASC-SLL 2024
Nishant Balepur, Abhilasha Ravichander, Rachel Rudinger
Selected: We discover LLMs score highly in MCQA benchmarks without the question, and analyze how

4. It's Not Easy Being Wrong: Large Language Models Struggle with Process of Elimination Reasoning [PDF]
ACL 2024 (Findings)
Nishant Balepur, Shramay Palta, Rachel Rudinger

3. Expository Text Generation: Imitate, Retrieve, Paraphrase [PDF]
EMNLP 2023, On UIUC's AI Reading List for Qualifying Exams, On Stanford's Reading List for CS224V
Nishant Balepur, Jie Huang, Kevin Chen-Chuan Chang

2. Text Fact Transfer [PDF]
EMNLP 2023
Nishant Balepur, Jie Huang, Kevin Chen-Chuan Chang

1. DynaMiTE: Discovering Explosive Topic Evolutions with User Guidance [PDF]
ACL 2023 (Findings)
Nishant Balepur*, Shivam Agarwal*, Karthik Ramanan, Susik Yoon, Diyi Yang, Jiawei Han

Preprints and Technical Reports

* denotes equal contribution, † denotes mentored student

6. Code Understanding is a Bottleneck for Coding Agents [PDF]
Under Review (ICLR)
Nishant Balepur, Kiran Tomlinson, Tobias Schnabel

5. (Im)Paired Programming: Coding Agents Improve Productivity but Harm Understanding [PDF]
Under Review (ARR)
Nishant Balepur, Connor Baumler, Valerie Chen, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber

4. DRACULA: Hunting for the Actions Users Want Deep Research Agents to Execute [PDF]
Under Review (ARR)
Nishant Balepur, ..., Rachel Rudinger, Eunsol Choi, Jordan Boyd-Graber, Doug Downey, Aakanksha Naik

3. Quick, Create a Distractor! Evaluating LLM Distractors for Multiple-Choice Benchmarks [PDF]
Under Review (ARR)
Atrey Desai†, **Nishant Balepur**, Rachel Rudinger

2. Position: Cognitive Alignment for Human Empowerment [PDF]
Under Review
Yoonjoo Lee, Philippe Laban, **Nishant Balepur**, Hariharan Subramonyam, Vera Liao

1. The Prompt Report: A Systematic Survey of Prompting Techniques [PDF]
Technical Report, Huggingface #1 Paper of the Day
Sander Schulhoff*, Michael Ilie*, **Nishant Balepur**, ..., Shyamal Anadkat, Alexander Hoyle, Phillip Resnik

Workshop Papers

* denotes equal contribution, † denotes mentored student

5. VibeJam: An Open Platform for User Studies on Agentic Vibe Coding [PDF]
   HCOMP 2026, North East AI Agents Day (Oral)
   Nishant Balepur, Connor Baumler, Valerie Chen, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber
4. BenchMarkerUI: An Interactive System for Auditing and Repairing Multiple-Choice Benchmarks
   HCOMP 2026
   Hyunjin Jane Oh†, **Nishant Balepur**, Jordan Boyd-Graber
3. Can They Dixit? Yes they Can! Dixit as a Playground for Multimodal Language Model Capabilities [PDF]
   Wordplay @ EMNLP 2025, Spotlight
   Nishant Balepur*, Dang Nguyen*, Dayeon Ki*
2. Language Models Generate Multiple-Choice Questions with Artifacts [Poster]
   MASC-SLL 2025
   Atrey Desai†, **Nishant Balepur**, Rachel Rudinger
1. Is Your Large Language Model Knowledgeable or a Choices-Only Cheater? [PDF]
   KnowLM @ ACL 2024
   Nishant Balepur, Rachel Rudinger

Invited Talks

Developing Question Answering Systems that Actually Help You — Virtual
CLIP Colloquium, University of Maryland — November 2026 (Upcoming)

Developing Question Answering Systems that Actually Help You — Virtual
Guest Lecture for CS 769 (Advanced NLP), University of Wisconsin–Madison — November 2026 (Upcoming)

Language Models Cheat on Multiple-Choice Benchmarks. Good! — Virtual
Machine Translation Unit, Fondazione Bruno Kessler (FBK) — September 2026

Understanding Understanding in Coding Agents — New York City
Pangram — August 2026

BenchMarker: A Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks — Virtual
CAMEL-AI — July 2026

Towards Helpful AI Agents — Virtual
FigX Lab, University of Maryland — March 2026

Aligning Models with Preferences is Misaligned with What Helps Users — New York, New York
ML2 Mixer, New York University — November 2025

Which of These Best Describes Multiple-Choice Evaluation with LLMs? — Virtual
Google Translate Reading Group — March 2025

Aligning LLMs to Answer Questions with Reasoning That Actually Helps You — London, U.K.
Imperial College, London — March 2025

Student Mentees

Matthew Shu (B.S. Yale → M.S. Yale → ML @ Brain Co.), 2023-2025, LLMs for Education, Alignment
Two papers at EMNLP 2024 (long, main); one paper at EMNLP 2025 (long, main)

Jerry He (HS Student → B.S. GTech), 2024-2025, LLMs for Education
Submitted Project on LLMs+Education for Marriott's Ridge High School Gifted and Talented Research Program

Michael Xie (M.S. UMD → Ph.D. JHU), 2025-2026, Personalized LLMs for Education, LLM Evaluation
One paper at ACL 2026, One paper in preparation

Atrey Desai (B.S. UMD), 2024-Present, LLM Evaluation
One paper at MASC-SLL 2025, Two papers at ACL 2026, One paper under review

Jane Oh (B.S. UMD), 2025-Present, LLM Evaluation, Human-AI Collaboration
One paper at ACL 2026, One paper at HCOMP 2026 (Demo).

Bhavya Rajasekaran (B.S. UMD), 2025, LLM Evaluation

One paper at ACL 2026

Service, Extracurriculars, and Teaching

Workshop Co-Organizer — UMD
Cognitive Alignment: Aligning AI with Human Cognitive Processes (Under Review, CHI) — 2026
The Science of Personas and LLM Personalization (Under Review, *ACL)

Conference Reviewer — UMD
Reviewer for: *ACL/ARR 2023–Present, IEEE TASLP 2024, TrustNLP 2024, SRW 2025 — 2022–Present
Great Reviewer Nomination: Apr, June (#6), Aug (#4), Oct (#2), Dec 2024, Feb 2025 ARR
Outstanding Reviewer Award: EMNLP 2024, EMNLP 2025
Ranked #4 Great Reviewer Overall: ARR

Conference Volunteer — UMD
ACL 2023, NAACL 2025 — 2023, 2025

Winter Storm LLM Workshop — UMD
Led a five-day workshop on LLM research for non-CS students in the language science center — Jan 2023

Visiting Student Day Volunteer — UMD
Volunteer for UMD's visiting student day for prospective graduate students — Mar 2024, 2025

Undergraduate TA for Linear Algebra with Computation Methods — UIUC
Led weekly office hours for 100+ students, created coding HW, and wrote outreach blogs — Aug 2020 - May 2023

Computer Science and Statistics Student Ambassador — UIUC
Mentor of ten incoming students and volunteer for computer science and statistics events — Aug 2022 - May 2023

SIGNLL — UIUC
President of Special Interest Group for Natural Language Learning — Aug 2020 - May 2021

Project: Code — UIUC
Co-founder of student organization to help new students build computer science projects — Aug 2019 - May 2021